

Auctions with Interdependent Values

- Bidder i gets “signal” X_i
- Ex-post value of bidder i :

$$V_i = v_i(X_1, X_2, \dots, X_N)$$

- This can also be a conditional expectation.
- “Pure” common value case:

$$V = v(X_1, X_2, \dots, X_N)$$

- Conditionally independent, unbiased signals case:

$$E[X_i | V = v] = v$$

- Also known as the “mineral rights” model.

Affiliation

- Strong form of correlation.
- If a subset of signals is high, it is more likely that the other signals are also large.
- Given $X_2, X_3, \dots, X_N$, i.e. all signals except X_1 , let $Y_1, Y_2, \dots, Y_{N-1}$ be the maximum, second highest, 3rd highest signals among signals aside from X_1
- Property 1: $E[\gamma(Y_1)|X_1 = x]$ is increasing in x
- Property 2: If $X_1, \dots, X_N$ are affiliated, so are $X_1, Y_1, \dots, Y_{N-1}$
- Property 3: X_1 and Y_1 affiliated implies the hazard rate

$$\frac{g_{Y_1|X_1}(y|x)}{G_{Y_1|X_1}(y|x)}$$

is increasing in x .

Symmetric model

- Bidders are symmetric, exchangeable

$$v_i(X_1, \dots, X_N) = v(X_i, X_{-i})$$

- Define

$$v(x, y) = E[V | X_1 = x, Y_1 = y]$$

expected value to bidder 1 if his signal is x , and the highest signal among competitors is y .

Second Price Auction

- Bidding strategy is given by

$$\beta^{SP}(x) = v(x, x)$$

- Suppose bidder 1 has value x . Value from winning if second-highest bidder had signal y , and used above strategy:

$$v(x, y) - \beta(y) = v(x, y) - v(y, y)$$

- This happens with probability $g_{Y_1|X_1}(y|x)$.
- So, expected payoff is

$$\Pi(b, x) = \int_0^{\beta^{-1}(b)} (v(x, y) - v(y, y)) g_{Y_1|X_1}(y|x) dy$$

- $v(x, y) - v(y, y) > 0$ for $y < x$, $v(x, y) - v(y, y) < 0$ for $y > x$, so marginal profit is zero when $x = \beta^{-1}(b)$, thus $\beta^{SP}(x) = v(x, x)$

The Winner's Curse

- Note:

$$v(x, x) = E[V|X_1 = x, Y_1 = x] \neq E[V|X_1 = x]$$

- “Naive” bidding: $E[V|X_1 = x]$
- Expected payment upon winning in second price auction

$$\begin{aligned} E(\beta^{SP}(Y_1)|X_1 = x, Y_1 < x) &= E(v(Y_1, Y_1)|X_1 = x, Y_1 < x) \\ &= \int_0^x v(y, y) d\left(\frac{G_{Y_1|X_1}(y|x)}{G_{Y_1|X_1}(x|x)}\right) \\ &< \int_0^x v(x, y) d\left(\frac{G_{Y_1|X_1}(y|x)}{G_{Y_1|X_1}(x|x)}\right) \\ &= E[V|X_1 = x, Y_1 < x] \end{aligned}$$

Expected payment upon winning less than expected value upon winning; thus not prey to winner's curse.

Empirical test for CV

- Exploit rational response to presence of the winner's curse
- In a second price auction, bid more cautiously if there are 100 competitors, as opposed to 2 (everything else being equal).

$$\begin{aligned}v(x, x, n) &= E[V|X_1 = x, X_2 = x, X_3 \leq x, \dots, X_{n-1} \leq x, X_n \leq x] \\&< E_{X_n} E[V|X_1 = x, X_2 = x, X_3 \leq x, \dots, X_{n-1} \leq x, X_n] \\&= E[V|X_1 = x, X_2 = x, X_3 \leq x, \dots, X_{n-1} \leq x] \\&= v(x, x, n - 1)\end{aligned}$$

where the first equality stems from the symmetry of the conditional expectation.

- Bid distribution in a second price auction with n bidders is stochastically dominated by the bid distribution with $n - 1$ bidders.

Identification with Fixed n

- Laffont and Vuong (1996), Athey and Haile (2002): without variation in n , we can't tell apart a common value model from an affiliated private value model.
- Why? Data gives us information only about the joint distribution of bids. This can be rationalized by a joint distribution of private values, which are correlated.
- Normalize signal distribution such that

$$v(x, x) = E[V|X_1 = x, Y_1 = x] = x$$

- Can not do this normalization if n varies.

First Price Auction

- Assume symmetric, increasing strategy, $\beta^{FP}(x)$
- Expected payoff from a bidder with signal x , but bids $\beta(z)$

$$\Pi(z, x) = \int_0^z (v(x, y) - \beta(z)) g_{Y_1|X_1}(y|x) dy$$

- First order condition wrt z :

$$(v(x, z) - \beta(z))g_{Y_1|X_1}(z|x) = \beta'(z)G_{Y_1|X_1}(z|x)$$

- At a symmetric equilibrium, $z = x$ for optimality, so:

$$(v(x, x) - \beta(x)) = \beta'(x) \frac{G_{Y_1|X_1}(x|x)}{g_{Y_1|X_1}(x|x)}$$

is the ODE characterizing equilibrium.

Revenue Comparison

- Second price auction yields more revenue than first-price auction.
- First-price auction bid strategy:

$$\beta^{FP}(x) = \int_0^x v(y, y) d \left(\exp \left(- \int_y^x \frac{g(t|t)}{G(t|t)} dt \right) \right)$$

- Compare with expected second-price payment (upon winning):

$$\begin{aligned} - \int_y^x \frac{g(t|t)}{G(t|t)} dt &\geq - \int_y^x \frac{g(t|x)}{G(t|x)} dt \\ &= \ln \left(\frac{G(y|x)}{G(x|x)} \right) \end{aligned}$$

Common Value Test for First-Price Auctions

- Haile-Hong-Shum: Can't compare the bids
- Competition effect vs. winner's curse effect
- Rewrite first-order condition in terms of bid distributions

$$v(x_i, x_i, n) = b_i + \frac{F_{M_i|B_i}(b_1|b_1)}{f_b(b_1|b_1)}$$

where $F_{M_i|B_i}(b_i|b_i)$ is the probability of the highest bid other than bidder i 's bid (M_i) being less than equal to b_i , conditional on $B_i = b_i$.

- Thus, we can estimate $v(x_1, x_1, n)$ from bids
- Use estimated quantities to do the stochastic dominance test.

Figure 2. Empirical Distributions of Pseudo-values
Scaled Sales

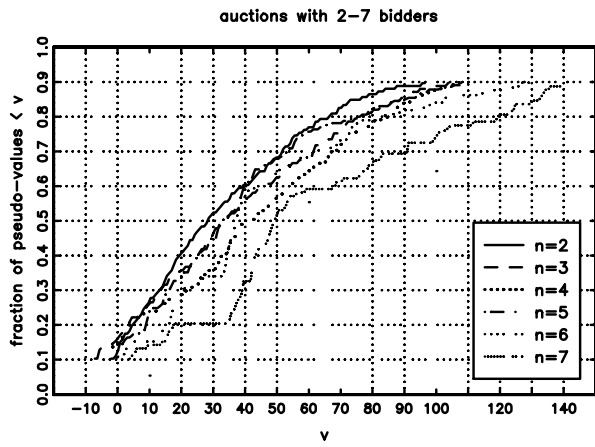
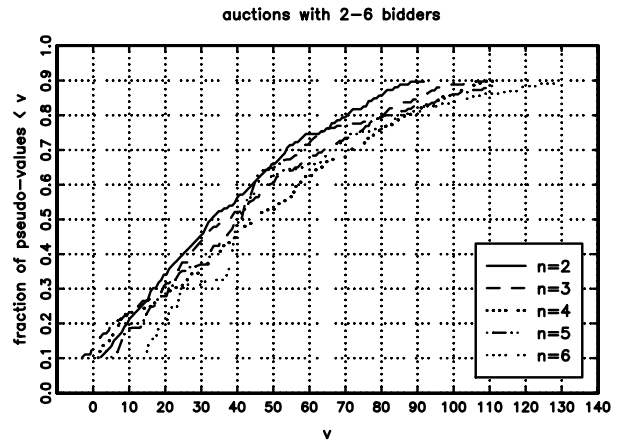
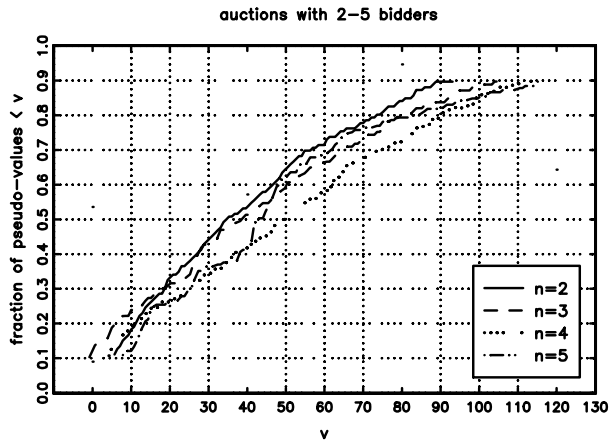
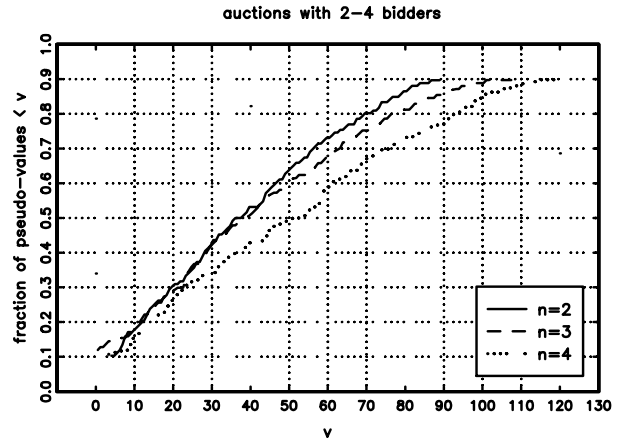
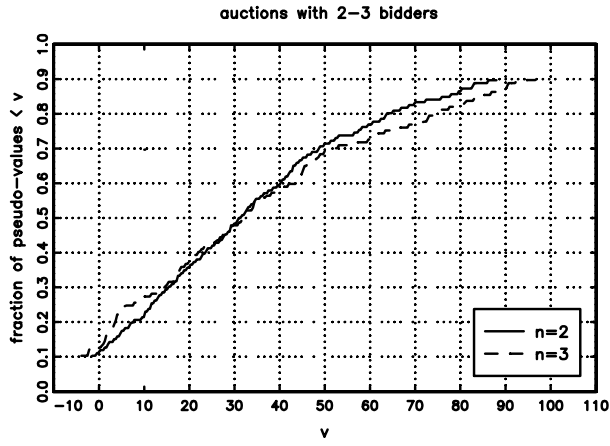


Figure 3. Empirical Distributions of Pseudo-values
Lumpsum Sales

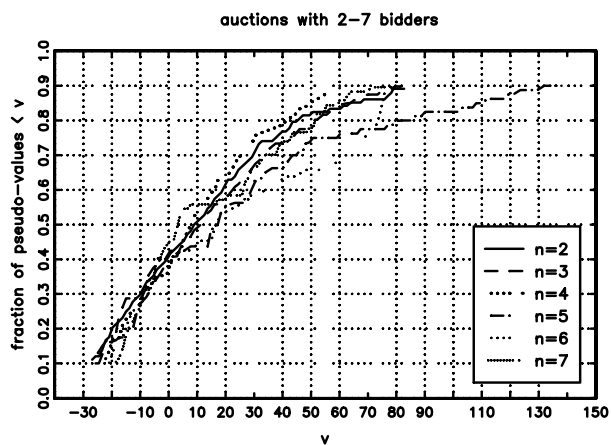
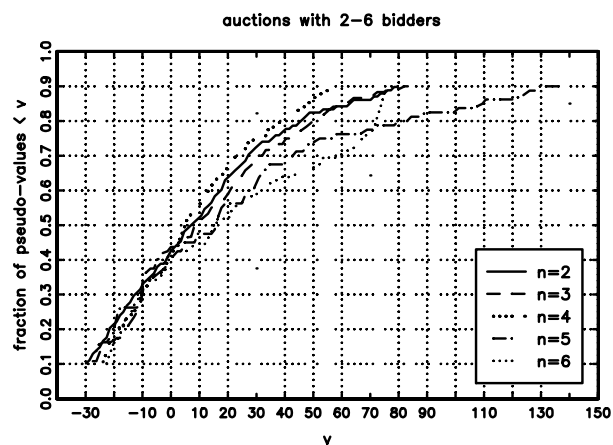
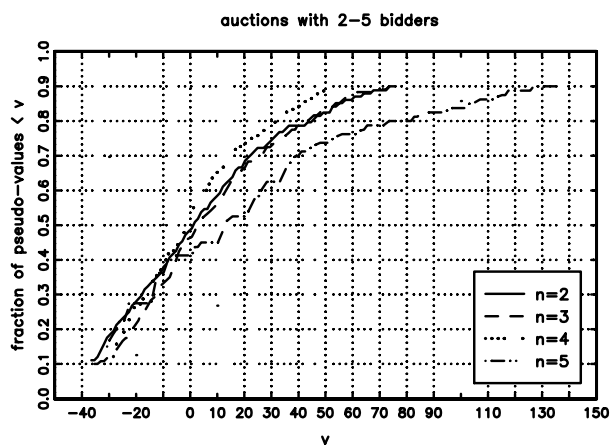
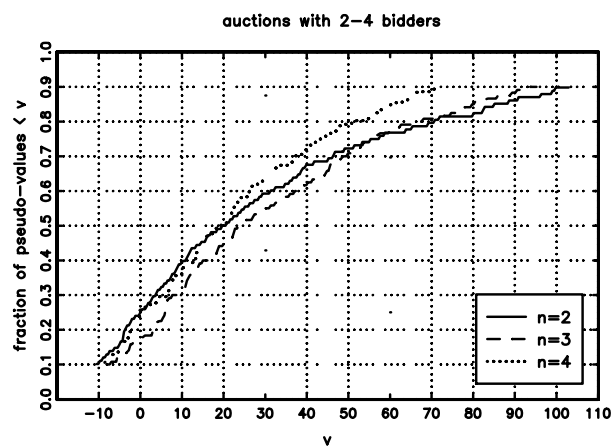
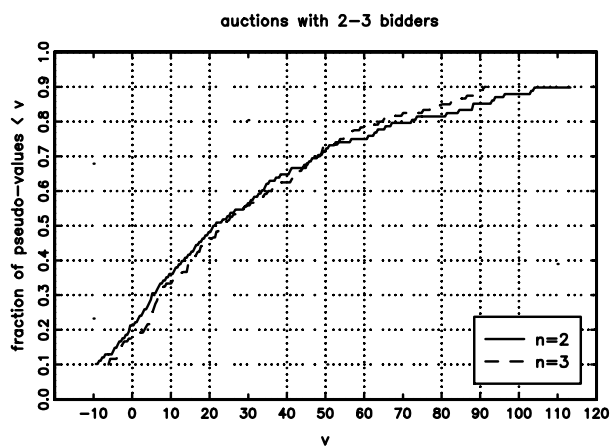


Figure 4. Empirical Distributions of Pseudo-values
Scaled Sales, Using Instrumental Variables

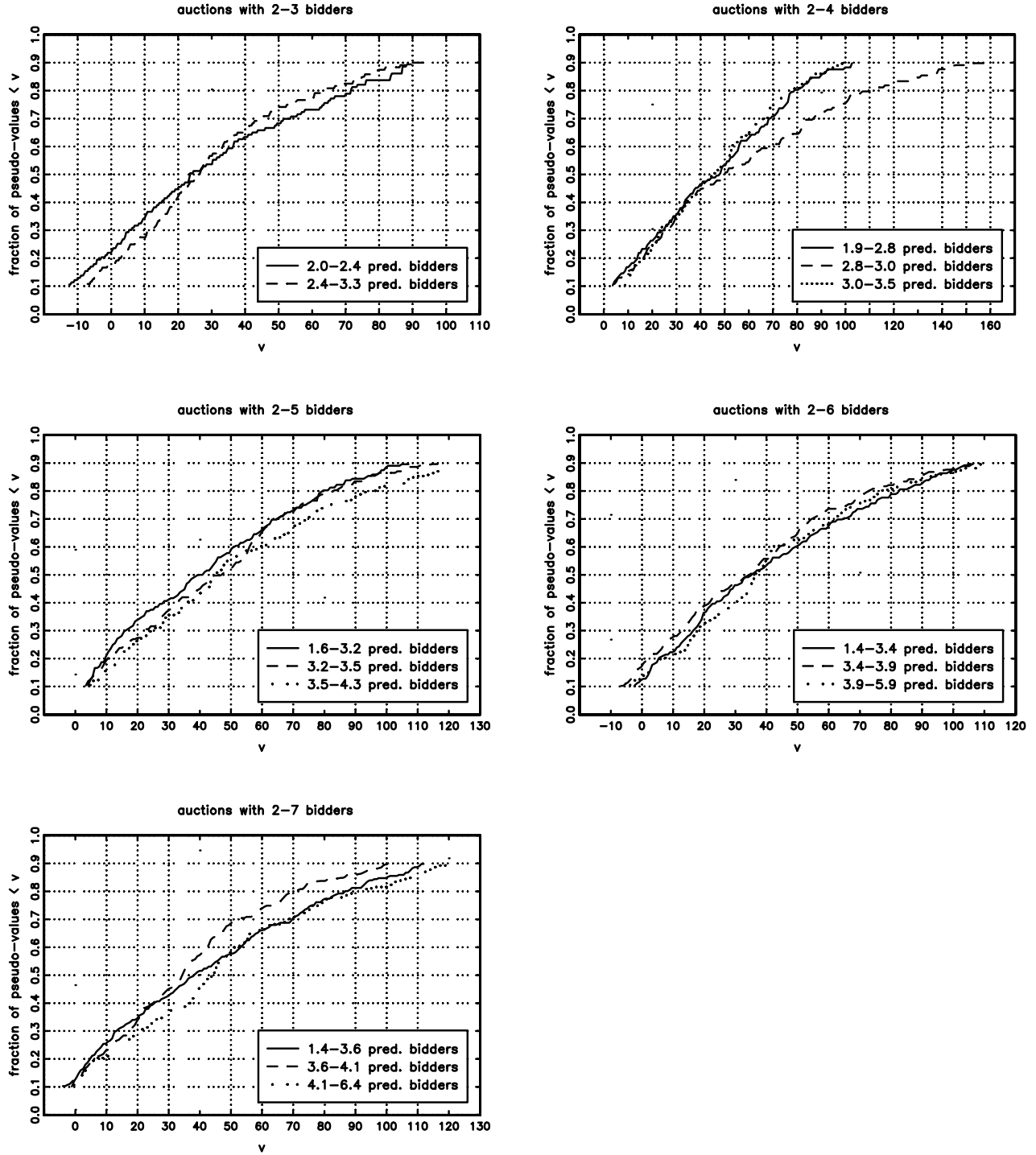
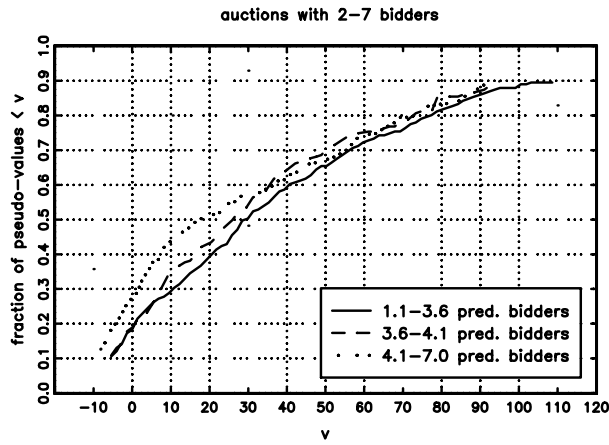
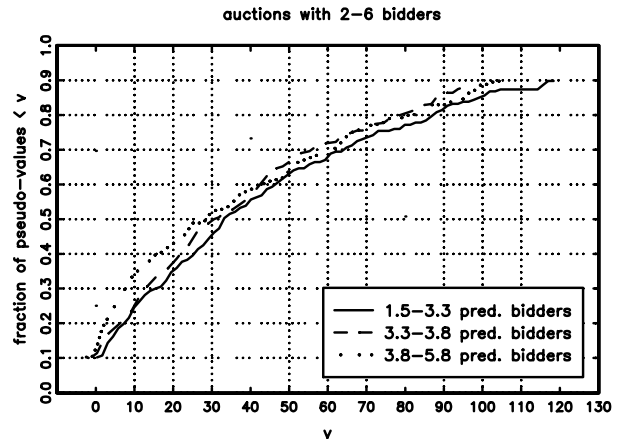
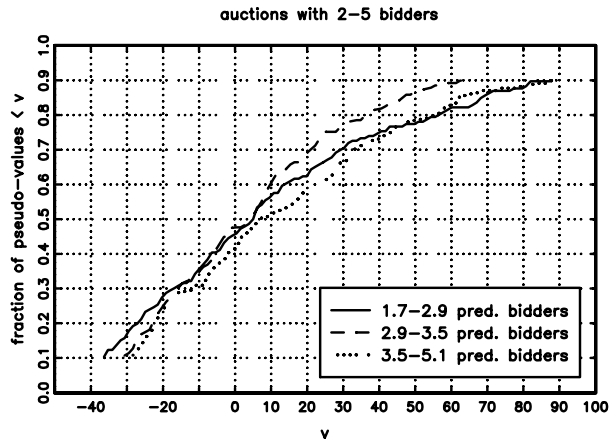
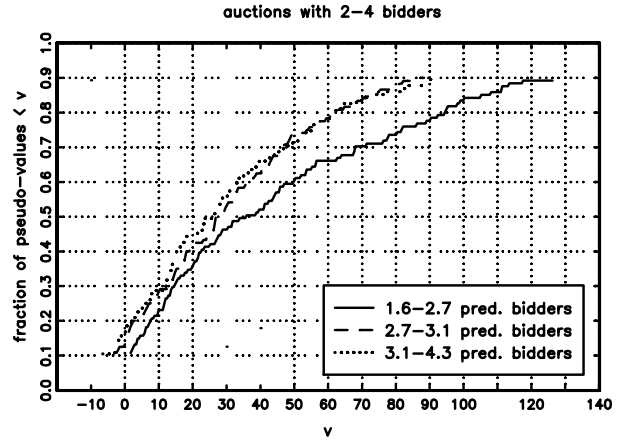
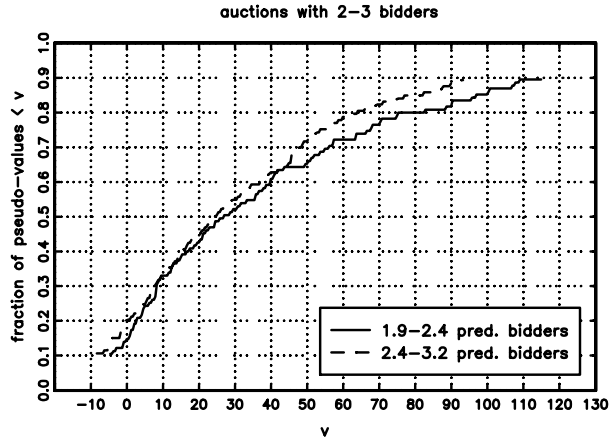


Figure 5. Empirical Distributions of Pseudo-values
Lumpsum Sales, Using Instrumental Variables



Ascending Auctions

- Identical to second-price auction with 2 bidders.
- Three bidders: last two remaining bidders learn from the drop-out point of the first person to drop out.
- Markov equilibrium: strategies conditioned on state
- State vector: Current price p , who dropped out, at what price
- Strategy: Bidder with signal x will drop out if

$$\beta^k(x, p_{k+1}, \dots, p_N) = p$$

- If nobody has dropped out, use $\beta^N(x)$, i.e. condition on own information only.

Characterization (4 bidders)

- Claim: There exists a symmetric equilibrium where

$$\beta^4(x) = E[V|X_1 = x, X_2 = x, X_3 = x, X_4 = x]$$

which, by affiliation, is increasing in x .

- If p_4 is the price at which the fourth bidder drops out:

$$\beta^3(x, p_4) = E[V|X_1 = x, X_2 = x, X_3 = x, X_4 = (\beta^4)^{-1}(p_4)]$$

- If p_3 is the price at which the third bidder drops out:

$$\beta^2(x, p_3, p_4) = E[V|X_1 = x, X_2 = x, X_3 = (\beta^3)^{-1}(p_3, p_4), X_4 = (\beta^4)^{-1}(p_4)]$$

Result (Milgrom and Weber 1982): The ascending auction revenue dominates the second price auction.

Drainage tract auction

- Two bidders, common value V
- One bidder knows V , the other draws signal X_2
- In equilibrium, uninformed bidder makes zero profit conditional on winning.

Common Value Auctions

- Everybody derives the same value from the object, v .
- Unfortunately, we do not know the value of the object for certain.
- Simplest setting: We all have the same information about the object.
- So we bid $E(v)$.
- Big problem with this: suppose there are two types of used cars, “lemons” (low v) and “non-lemons” (high v).

- "Non-lemons" will be driven out of the market – but then, nobody will want to bid!
- This is the classic adverse selection problem.
- Now let's consider a setting where there are two types of bidders
- Those who know what v is, and those who don't.
- Suppose there are two guys, one who knows whether v is high or low, the other doesn't. We run a second price auction. Let's say v is high with probability π .

- Claim: the guy who knows, bids his valuation. The guy who doesn't, bids the low value of v .
- Corollary 1: The uninformed guy makes zero profit upon winning the auction.
- Corollary 2: The uninformed guy bids less than his expected value.
- Corollary 3: the expected revenue of the seller is less than $E(v)$.

Real-life examples of the above phenomenon:

1. Oil drilling
2. Sniping: when would an informed bidder bid on eBay?
3. IPO discount

The following is from Henricks and Porter (1988), in the American Economic Review.

They look at auctions for oil tracts, where:

- Some bidders own neighboring tracts
- Some bidders don't own neighboring tracts
- What may be an advantage of bidders who own neighboring tracts?
- What might we expect about the profits of neighbor vs. non-neighbor bidders?

TABLE 3—SAMPLE STATISTICS ON TRACTS WON BY EACH TYPE OF FIRM^a

	Wins by Neighbor Firms		Wins by Non-Neighbor Firms		
	A	Total	B	C	Total
No. of Tracts	35	59	19	36	55
No. of Tracts Drilled	23	47	18	33	51
No. of Productive Tracts	16	36	12	19	31
Average Winning Bid	3.28 (0.56)	6.04 (2.00)	2.15 (0.67)	6.30 (1.31)	4.87 (0.92)
Average Gross Profits	10.05 (3.91)	12.75 (3.21)	-0.54 (0.47)	7.08 (2.95)	4.45 (1.99)
Average Net Profits	6.76 (3.02)	6.71 (2.69)	-2.69 (0.86)	0.78 (2.64)	-0.42 (1.76)

^aDollar figures are in millions of \$1972. The numbers in parentheses are the standard deviations of the sample means. Column A refers to tracts which received no non-neighbor firm bid, column B refers to tracts which received no neighbor bid, and column C to those in which a neighbor firm bid, but a non-neighbor firm won the tract.

neighbor firms. The number of tracts which received no neighbor bid is 19, and the number of tracts which received no non-neighbor bid is 35. Therefore, at least one neighbor firm participated in 83 percent of the auctions, and at least one non-neighbor firm participated in 68 percent of the auctions. This is consistent with the theoretical model.

Table 3 gives sample statistics on the tracts won by each type of firm. Column A refers to tracts which received no non-neighbor firm bid, column B refers to tracts which received no neighbor bid, and column C to those in which a neighbor firm bid, but a non-neighbor firm won the tract.

The evidence is consistent with the model. The neighbor firm won 62 percent of the tracts that it bid on. As we calculated in our previous paper, its share of the tract value was approximately 44 percent, which was considerably higher than the 23 percent average firm share on wildcat tracts. The average net profit of non-neighbor winners was virtually zero. It was positive on tracts which received a neighbor bid, and it was significantly negative on tracts which received no neighbor bid. By contrast, the participation decisions of the non-neighbor firms had no effect on the earnings of neighbor firms. Based on these return figures, it appears as if the neighbor firm was better able to identify which drainage tracts were more likely to contain oil, and was able to

exploit this knowledge to obtain above average profits.

We found no evidence of a mass point at the reservation price in the distribution of bids of the neighbor firm. One possible explanation for this result is that firms were afraid that the government would reject reservation price bids. Recall that the government rejected the high bid on 25 drainage tracts.³

The existence of a positive reservation price provides an explanation for why non-neighbor firms drilled tracts on which no neighbor firm bid. The lack of participation by the neighbor firm implies that its expectation of net profit is less than R . Since R is positive, and the bid is a sunk cost, it may still have been rational for the non-neighbor firm to drill its lease. Drilling outcomes were not inconsistent with this belief, since the average gross profit of tracts which received no neighbor bid was not significantly different from zero at conventional confidence levels.

An indirect test of our assumption that neighbor firms coordinated their bidding decisions is to compare the bidding behavior of neighbor firms and their net profits on single neighbor tracts to that on multiple neighbor

³We study the issue of a random reservation price, and its empirical implications, in a subsequent paper.

The following is from Ockenfels and Roth (2002), in the American Economic Review.

They look at the frequency of “sniping” in eBay vs. Amazon, where:

- eBay has a firm ending time for each auction
- Amazon has a “going-going-gone” rule.
- Where do you expect bidders to snipe?
- They also look across antiques vs. computer auctions.
- For which items do you expect more sniping?

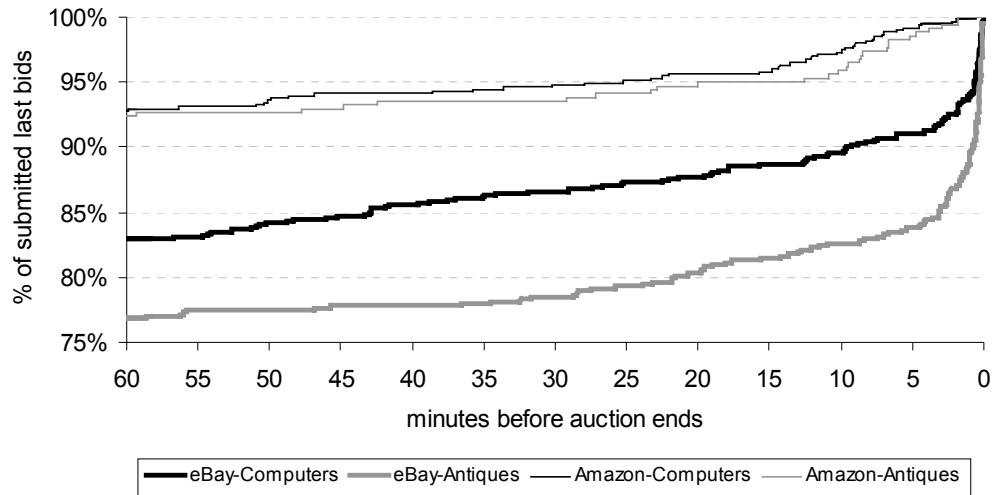


Figure 1a—Cumulative distributions over time of bidders' last bids

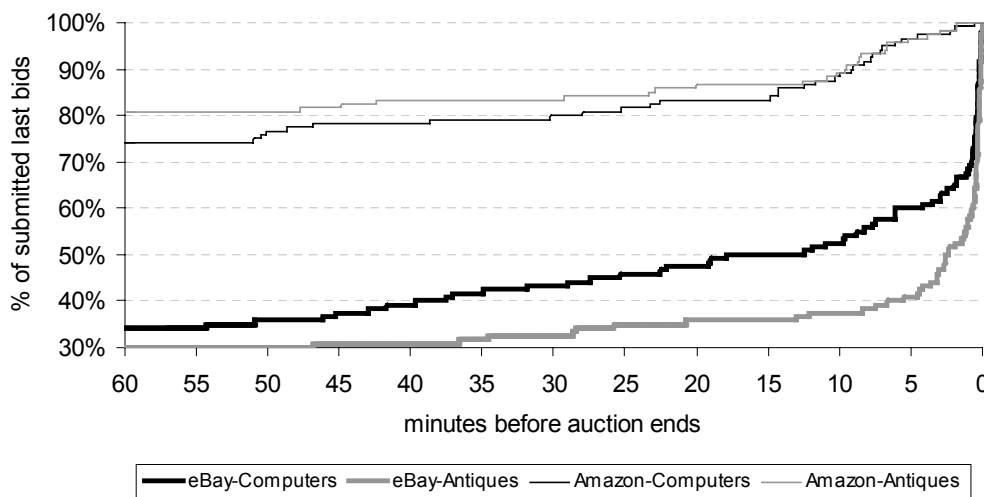


Figure 1b—Cumulative distributions over time of auctions' last bids

Figure 1a reveals that 20 percent of all last bids on eBay compared to 7 percent of all last bids on Amazon were submitted in the last hour. Figure 1b shows that in more than two thirds of all eBay auctions, at least one bidder is still active in the last hour, while this is only true for about one quarter of all Amazon auctions. Furthermore, the figures reveal that, on eBay, a considerable share of bidders submit their bid in the last five minutes (9 percent in Computers and 16 percent in Antiques), while only few bids come in equally late on Amazon (about 1

Of course, there can also be non-strategic reasons why bidders bid late, some of which are listed in Table 1. These non-strategic reasons, however, should be relatively unaffected by the difference in rules between eBay and Amazon. (The hypotheses are not mutually exclusive; they could each be contributory causes of late bidding.)

	Hypotheses	Predicted contribution to late bidding
Strategic hypotheses	• <i>Rational response to naïve English auction behavior or to shill bidders</i>: bidders bid late to avoid bidding wars with incremental bidders. • <i>Collusive equilibrium</i>: bidders bid late to avoid bidding wars with other like-minded bidders. • <i>Informed bidders protecting their information</i>: e.g. late bidding by experts/dealers.	All three strategic hypotheses suggest more late bidding on eBay than on Amazon, with a bigger effect for more experienced bidders. Plus (via the third point) more late bidding in categories in which expertise is important than in categories in which it is not.
Non-strategic hypotheses	Bidders bid late because ... • of procrastination; • search engines present soon-to-expire auctions first; • of a desire to retain flexibility to bid on other auctions offering the same item; • they remain unaware of the proxy bidding system; • of an increase in the willingness to pay over time caused by, e.g., an endowment effect; or because • bidders don't like to leave bids "hanging."	No difference between eBay and Amazon.

Table 1—Hypotheses about the causes of late bidding

The strategic differences between eBay style, hard close auctions and Amazon style, automatic extension auctions suggest that the hypotheses about the causes of late bidding can be investigated by examining the timing of bids on eBay and Amazon. So, we compare the timing

(<http://www.esnipe.com/faq.asp>, 2000). In fact, esnipe.com recently started to publish statistics on success rates, time to place bids and hourly trends based on an average of more than 4200 bids per day (esnipe.com/stats.asp, 2000). While the time it takes to place a bid varies considerably over weekdays and hours, on average 4.5 percent of esnipe's late bids failed to be successfully transmitted in September 2000. (Esnipe was sold on eBay in an auction

The Winner's Curse:

- In a common value auction, the winner of the auction has the highest estimate of the value of the good.
- Hence, the winner runs into the danger of overestimating the value of the good.
- The rational response by bidders is to reduce their bids by taking into account the fact that they win conditional on being the most optimistic.
- Claim 1: Bidders will act more conservatively when they face more competitors

- Claim 2: Bidders will act more conservatively when they face higher uncertainty about the value of the object.
- Both of these claims have found some backing in empirical tests.

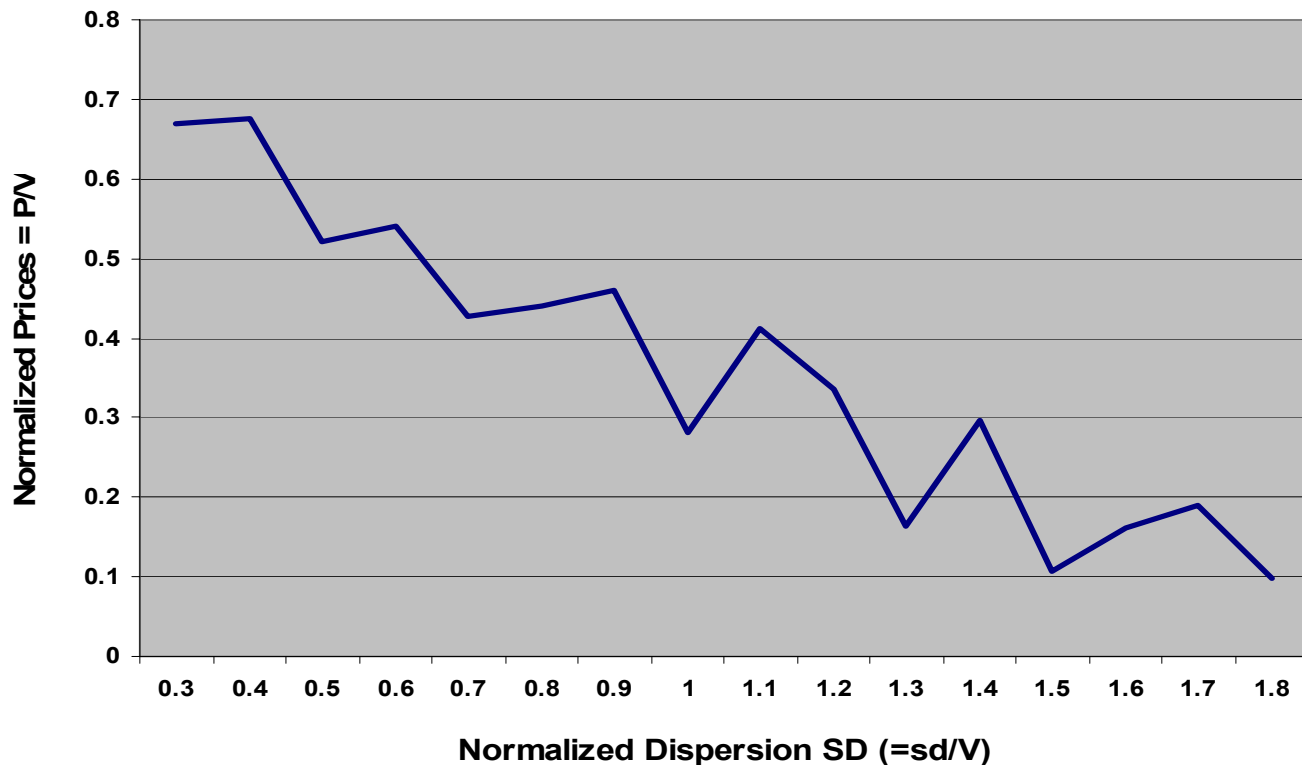
The following figure is from Yin (2003), referenced in Bajari and Hortacsu (2003)

Yin looks at the prices in eBay computer auctions, as a function of the dispersion of appraisals for the auctions listings, where the appraisals are made by survey participants.

- Do you expect price to go up or down as uncertainty increases?

and therefore sellers have a strong incentive to communicate what they are selling by designing a clear web page to reduce bidder uncertainty.

Figure 2: Bidder Uncertainty and Winning Bids.



A potential limitation to Yin's analysis is that the variance of x for survey respondents may be an imperfect measure of the variance of x for the real bidders. The survey respondents' reservation values are roughly twice the final sales price on average. Perhaps they are not well informed about used computer prices. On the other hand, if the winner's curse is a strong possibility, it may be an equilibrium for a bidder to shade his bid to be one half of his estimate x .

Is There a Winner's Curse on eBay?

Conditional on their caveats, all three papers surveyed in this section suggest that the "winner's curse" is an important concern in the eBay markets analyzed (baseball cards, collectible coins, used computers). Two of the three papers, Bajari and Hortacsu (2003) and Yin (2003), find evidence that bidders strategically respond to the presence of